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Crossing Shannon's Boundary: The FATIMA Semantic Integrity Engine and the Long-Term Architecture of Positive Fractal Growth

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Abstract

Every data-integrity system ever built operates below what this paper terms Shannon's boundary: it can verify that bits are preserved, but it cannot verify that *meaning* is preserved. A file whose every byte survives transmission intact has syntactic integrity. But a document whose conclusion is negated—"Allah is not One" replacing "Allah is One"—while every other byte remains identical has suffered a catastrophic meaning failure that no hash, no checksum, and no error-correcting code can detect, because the bits are exactly what the adversary intended them to be. FATIMA—Fitrah-Aligned Tawhidic Integrity for Meaning and Authenticity—was introduced in [2] as a specification for a holographic data format that crosses this boundary. The companion paper [3] documented the evolution of the encoder through v5.4.0, acknowledging in its honest caveats that the encoder "currently operates below that boundary—it protects bits, not meaning." This paper documents v6.0: the Semantic Integrity Engine that begins the crossing. Seven semantic levels—lexical, syntactic, propositional, relational, rhetorical, thematic, and intentional—run as a parallel witness chain alongside the 154 syntactic layers, producing a dual-witness seal that answers two questions simultaneously: "Are the bits intact?" and "Is the meaning intact?" We present the architecture, the seven-level computation, the dual-witness integration, and the results of three verification scenarios that demonstrate the boundary crossing. We then present the long-term roadmap for positive fractal growth: the architectural trajectory from the current v6.0 through a projected v10.0, in which the encoder grows not by accretion but by self-similar extension at every scale—each new capability encoding the same theological constants (7, 11, $154 = 2 \times 7 \times 11$) at a higher order of protection.

Keywords: FATIMA, Shannon's boundary, semantic integrity, meaning preservation, dual witness, fractal growth, Al-Asma ul-Husna, positive fractal architecture, Bismillah

1. Introduction: The Problem Below the Boundary

Claude Shannon's 1948 paper "A Mathematical Theory of Communication" [1] established the theoretical framework for all modern data integrity. Shannon's channel coding theorem guarantees that for any noisy channel, there exists a coding scheme that reduces the probability of transmission error to an arbitrarily small value. Every system built on this foundation—from TCP checksums to SHA-256 to the HMAC witness sidecars in FATIMA's own 154-layer architecture—operates on the same assumption: *integrity means preserving bits.*

This assumption has been spectacularly successful. The entire infrastructure of the modern internet depends on it. But it contains a gap so fundamental that it is invisible: bit-preservation is necessary but not sufficient for meaning-preservation.

Consider a sealed document containing the shahada: *La ilaha illa Allah*—there is no god but Allah. The FATIMA v5.4.0 encoder seals this document with 154 layers of Bismillah-anchored SHA-256 hashing across eleven heavens. Any single-bit change to any byte in the file will be

detected. But suppose an adversary produces a *new* file—not a corruption of the old one, but a fresh document—containing *La ilaha illa al-shaytan*. This new file will, of course, have a completely different hash. The 154-layer witness will report that the file has changed. But it will report the change the same way it reports a reformatted paragraph, a corrected typo, or an updated date. The syntactic witness has no concept of the catastrophic semantic inversion that has occurred. It sees bits, not meaning.

This is Shannon's boundary. Below it, we protect symbols. Above it, we protect what the symbols *mean*. The entire history of data integrity operates below this boundary. FATIMA v6.0 begins the crossing.

2. Theological Grounding: Why Meaning Must Be Protected

The Qur'an does not merely record information; it protects meaning. The divine promise in Surah Al-Hijr (15:9)—*"Indeed, it is We who sent down the Reminder, and indeed, We will be its Guardian"*—is not a promise to preserve bits. The Qur'an has been transmitted through multiple *qira'at* (recitation styles), each with minor phonetic variations, yet the *meaning* is preserved across all of them. This is semantic integrity at the divine scale: the surface representation varies while the meaning remains invariant.

Surah Fussilat (41:42) makes the protection explicit: *"Falsehood cannot approach it from before it or from behind it; a revelation from One who is Wise and Praiseworthy."* This verse does not say that the *letters* cannot be approached by falsehood—a physical manuscript can be burned, a digital file can be corrupted. It says that *falsehood* cannot approach *it*—the meaning, the content, the *dhikr* (reminder) itself. The protection is semantic, not syntactic.

The five FATIMA properties identified in [2]—holographic redundancy, fitrah-alignment, tawhidic unity, meaning-integrity, and authenticity verification via structural coherence—demand a system that can distinguish between surface change and meaning change. The first four versions of the encoder could not. Version 6.0 begins to.

The number of semantic levels—seven—is not arbitrary. The Qur'an establishes seven as the muhkam (clear, self-evident) constant: *"[He] who created seven heavens in layers"* (67:3); *"Do you not consider how Allah has created seven heavens in layers?"* (71:15). If the syntactic architecture embeds both 7 and 11 in $154 = 2 \times 7 \times 11$, the semantic architecture isolates the muhkam constant: seven levels of meaning, each independently witnessed. The full dual-witness system thus encodes:

- 154 syntactic layers ($2 \times 7 \times 11$): the bits across all scales of creation.
- 7 semantic levels: the meaning at every depth of interpretation.
- $154 + 7 = 161$ total witness dimensions.
- The dual-witness seal: a single SHA-256 hash that unifies both chains.

3. The Seven Semantic Levels

The Semantic Integrity Engine computes seven levels of meaning-witness, ordered from surface to depth. Each level extracts a different aspect of the document's meaning, normalizes it to a canonical form, and seals it with a Bismillah-anchored SHA-256 hash that cycles through Al-Asma ul-Husna—the same Divine Name cycling used by the 154 syntactic layers, ensuring that the semantic chain participates in the same theological witness.

3.1. Level 0: Lexical (Word Inventory)

The lexical level answers: *what words does this document use?*

The engine tokenizes the document, strips stopwords (articles, prepositions, auxiliaries), and computes the sorted set of unique content words. The hash of this sorted set is invariant to word order, sentence structure, and formatting—only the *vocabulary* matters. A document that

says "The cat sat on the mat" and a document that says "On the mat, the cat sat" produce the same Level 0 seal. But a document that says "The dog sat on the mat" does not.

This is the shallowest semantic level: two documents share a lexical seal only if they use exactly the same content words, regardless of arrangement.

3.2. Level 1: Syntactic (Sentence Structure)

The syntactic semantic level answers: *how are the sentences structured?*

For each sentence, the engine computes a structural pattern: the sequence of word lengths. "The quick brown fox jumps" becomes the pattern `[3,5,5,3,5]`. The hash of these patterns captures the *rhythm* of the prose—its cadence, its sentence complexity, its variation. A paraphrase that replaces words with synonyms of different lengths will change this level; a replacement with synonyms of the same length will not.

This level sits between surface and meaning: it captures grammatical structure without reference to semantics.

3.3. Level 2: Propositional (Assertions)

The propositional level answers: *what does this document claim?*

The engine extracts normalized propositions from each sentence. The current implementation uses a lightweight approximation: for each sentence, it extracts the first three content words as a (subject, predicate, object) triple, normalizes each word to lowercase, and sorts the triple alphabetically. This normalization means "God created the heavens" and "The heavens were created by God" produce the same propositional hash—different surface forms of the same assertion.

Critically, this level is *negation-sensitive*: the engine preserves negation words during tokenization. "God created the heavens" and "God did *not* create the heavens" produce different propositional hashes. This is the feature that distinguishes semantic integrity from syntactic integrity: the ability to detect meaning inversion.

3.4. Level 3: Relational (Entity Relations)

The relational level answers: *how are the named entities connected?*

The engine identifies capitalized words as entities and records which entities co-occur within the same sentence. The hash captures the *relationship graph* of the document: who is mentioned alongside whom, which concepts are linked, which topics are discussed together. A document about "Allah and His Messenger" produces a different relational seal than a document about "Allah and His creation," even if the rest of the text is identical.

3.5. Level 4: Rhetorical (Argument Flow)

The rhetorical level answers: *how does the document argue?*

The engine tracks logical connectors—*therefore*, *because*, *however*, *despite*, *unless*, *if…then*—and records both their sequence and their frequency distribution. The hash captures the *argumentative structure*: whether the document proceeds by deduction, concession, contrast, or accumulation. Two documents that reach the same conclusion through different reasoning patterns will produce different rhetorical seals.

3.6. Level 5: Thematic (Topic Distribution)

The thematic level answers: *what is this document about?*

The engine computes a lightweight topic model using word co-occurrence in fixed-size windows. The content words are divided into $n = 7$ buckets (the muhkam constant again), and the

most frequent words in each bucket are hashed together. The result is a topical fingerprint: a document about "divine protection and file integrity" produces a different thematic seal than a document about "market analysis and portfolio optimization," even if they share much of their vocabulary.

The choice of 7 topic buckets is theologically grounded: the seven heavens in Islamic cosmology represent seven domains of creation, and the seven semantic levels represent seven domains of meaning. The thematic level subdivides the document's content into seven topical heavens.

3.7. Level 6: Intentional (Speech Acts)

The intentional level answers: *what is this document trying to do?*

The engine classifies sentences by their illocutionary force—the speech-act framework of Austin and Searle applied computationally. Five categories are tracked:

- **Assertive**: claims, conclusions, observations ("This proves…")
- **Directive**: commands, requests, requirements ("You must…")
- **Commissive**: promises, commitments, pledges ("We commit to…")
- **Expressive**: praise, gratitude, apology ("Alhamdulillah…")
- **Declarative**: definitions, verdicts, announcements ("We declare…")

The hash captures the *purpose distribution* of the document: a legal contract (high directive, high commissive) produces a different intentional seal than a research paper (high assertive, low directive), even if both discuss the same subject matter. The engine also tracks negated speech acts—assertions that negate other assertions—as a separate count, providing sensitivity to documents that argue *against* positions rather than *for* them.

3.8. The Semantic Master Seal

After all seven levels are computed, the engine chains their seals into a single Semantic Master Seal—analogous to the Sidrat al-Muntaha that caps the 154 syntactic layers. The master seal is a SHA-256 hash computed as:

```
BISMILLAH + "SEMANTIC_MASTER_WITNESS"
  + Level_0_seal + Level_1_seal + ...
  + Level_6_seal + BISMILLAH
```

The Bismillah anchoring at both ends mirrors the syntactic chain: every computation begins and ends with the invocation of Allah's protection. The semantic master seal is a single 256-bit value that represents the *meaning* of the document across all seven levels.

4. The Dual-Witness Architecture

FATIMA v6.0 does not replace the 154 syntactic layers with the 7 semantic levels. It runs both as parallel witness chains, then unifies them in a dual-witness seal. This design is grounded in the principle of *tawhidic duality*: the syntactic and semantic witnesses are distinct aspects of a single truth, just as the *zahir* (manifest) and *batin* (hidden) aspects of revelation are distinct but unified.

The dual-witness seal is computed as:

```
BISMILLAH + "DUAL_WITNESS:SYNTACTIC+SEMANTIC"
  + syntactic_master_witness
  + semantic_master_seal
  + BISMILLAH
```

This produces a single SHA-256 hash that can only be reproduced if *both* the bits and the meaning are intact. If either chain breaks, the dual seal breaks. The dual-witness seal is the

encoder's first artifact that operates above Shannon's boundary: it answers both "are the bits intact?" and "is the meaning intact?" in a single verification.

The sidecar witness file (the `.witness.bismillah` JSON) now contains three sections:

1. **heavens**: the 11-heaven, 154-layer syntactic witness (unchanged from v5.4.0).
2. **semantic_witness**: the 7-level semantic witness with per-level seals, divine names, and metadata.
3. **dual_witness_seal**: the unified seal bridging both chains.

Backward compatibility is preserved: any sidecar produced by v5.4.0 can still be verified by v6.0's syntactic chain. The semantic witness is additive, not destructive.

5. Verification: Three Scenarios Crossing the Boundary

To demonstrate the boundary crossing, we tested the semantic verification engine against the Evolution Paper [3] itself under three adversarial scenarios.

5.1. Scenario 1: Surface Modification (Meaning Preserved)

The original document was modified by changing whitespace, adjusting formatting, and altering punctuation—changes that modify bytes but preserve meaning. Result:

Level	Preserved?	Note
0. Lexical	✓	Same vocabulary
1. Syntactic	✓	Same sentence patterns
2. Propositional	✓	Same assertions
3. Relational	✓	Same entity relations
4. Rhetorical	✓	Same argument flow
5. Thematic	✓	Same topic distribution
6. Intentional	✓	Same speech acts
Result	**7/7**	**BOUNDARY CROSSED**

The syntactic witness reports failure (bits changed). The semantic witness reports success (meaning preserved). The dual system correctly identifies this as a surface modification—*above* Shannon's boundary.

5.2. Scenario 2: Meaning Alteration

Key assertions were changed: conclusions were reversed, numbers were altered, and technical claims were modified. Result:

Level	Preserved?	Note
0. Lexical	×	Different vocabulary
1. Syntactic	×	Different structure
2. Propositional	×	Different assertions
3. Relational	×	Different entity relations
4. Rhetorical	✓	Similar argument flow
5. Thematic	✓	Similar topics
6. Intentional	✓	Similar purpose
Result	**3/7**	**MEANING ALTERED**

The system correctly identifies that while the document retains its general structure and purpose (rhetorical, thematic, and intentional levels intact), its specific claims have been changed. The diagnosis reports: "Vocabulary changed," "Propositions changed," "Entity relationships changed."

5.3. Scenario 3: Negation Insertion

The most theologically critical test: key assertions were negated without other changes. "FATIMA *is* a holographic data format" becomes "FATIMA is *not* a holographic data format." Result:

Level	Preserved?	Note
0. Lexical	×	"not" added
1. Syntactic	×	Pattern lengths changed
2. Propositional	×	Assertions negated
3. Relational	×	Different co-occurrence
4. Rhetorical	×	Connector context shifted
5. Thematic	✓	Same broad topics
6. Intentional	✓	Still assertive in form
Result	**2/7**	**MEANING ALTERED**

The negation test is the quintessential Shannon's-boundary test. The syntactic change is minimal—a few "not" insertions across thousands of words. A traditional hash would detect the bit-level change but report it identically to a formatting change. The semantic engine correctly identifies this as a meaning-critical alteration, with the propositional and relational levels breaking precisely because the *assertions* have been inverted.

6. Honest Assessment: What v6.0 Does and Does Not Achieve

The Semantic Integrity Engine is a first step across Shannon's boundary, not the crossing itself. Several limitations must be acknowledged:

1. **Lightweight NLP.** The engine uses pure Python with no external NLP dependencies. This is an intentional design choice—the encoder must run anywhere, without requiring TensorFlow, spaCy, or NLTK—but it limits the depth of semantic analysis. The propositional extraction uses a heuristic (first three content words) rather than a parser. The topic model uses word co-occurrence rather than LDA or neural embeddings.
2. **Language sensitivity.** The stopword list, sentence detection, and tokenization are optimized for English. Documents in Arabic, Urdu, or other languages will produce semantic witnesses, but with reduced accuracy. Extending the engine to full multilingual support is future work.
3. **Threshold sensitivity.** The verification threshold (default 0.5, meaning $\geq 4/7$ levels must match) is a parameter, not a proof. Different applications may require different thresholds. A theological text might demand 7/7 (no meaning change at any level), while a software README might tolerate 4/7 (same purpose and structure, different specifics).
4. **No semantic distance.** The current engine reports binary match/no-match per level. It does not compute *how much* meaning has changed—only *whether* it has. A future version should compute semantic distance, not just semantic equivalence.
5. **Adversarial robustness.** A sophisticated adversary who understands the seven levels could craft a document that inverts meaning while preserving all seven seals—for example, by using exactly the same vocabulary in exactly the same patterns but with meaning-altering context that the heuristic extractors cannot detect. The engine raises the bar from bit-level to meaning-level, but it does not make the bar unbreakable.

7. The Roadmap: Positive Fractal Growth

The evolution of the FATIMA encoder from v1.0 through v6.0 reveals a pattern: each version responds to a limitation exposed by the previous version's deployment, and each response encodes the same theological constants at a higher order. This is not accidental growth; it is *positive fractal growth*—self-similar extension at every scale, where the growth itself is structured by the same principles that structure the architecture.

The term "positive fractal" distinguishes this trajectory from negative fractality (self-similar degradation, as in the Hurst exponent's identification of long-range memory in corruption patterns [4]). Positive fractality means that the system's growth replicates its foundational structure at each new scale of capability:

- 7 appears in the muhkam constant, in the 7 semantic levels, and in the 14 layers per heaven ($14 = 2 \times 7$).
- 11 appears in the 11 heavens, in the dimensional constant $H(11) = 1.0157$, and in the architecture's volumetric cardinality.
- $154 = 2 \times 7 \times 11$ appears in the total syntactic layers, encoding both constants simultaneously.
- 99 appears in Al-Asma ul-Husna, the Divine Names that cycle through every layer and level.

Positive fractal growth means that every new capability added to the encoder must encode these constants at its own scale, not merely sit alongside them. The v6.0 semantic engine achieves this: 7 levels, Bismillah-anchored, Divine-Name-cycling, unified with the 154-layer chain through the dual-witness seal.

The roadmap extends this principle across four projected versions:

7.1. v7.0: Temporal Integrity (The Living Witness)

Current limitation: v6.0 verifies meaning at a single point in time. It cannot distinguish between a document that was legitimately updated and one that was maliciously replaced.

Proposed response: a temporal witness chain that records the *evolution* of both syntactic and semantic state across time. Each sealing operation appends to a Merkle tree whose leaves are dual-witness seals, producing a verifiable history of the document's meaning over time. The tree structure would encode 11 temporal epochs per document lifecycle and 7 semantic checkpoints per epoch—77 temporal witnesses encoding both divine constants at the temporal scale.

Theological grounding: *"Indeed, We have made it a Qur'an in Arabic so that you might understand"* (43:3). The Qur'an's meaning is preserved not only spatially (across copies) but temporally (across centuries). The temporal witness extends FATIMA's protection to the time dimension.

7.2. v8.0: Relational Integrity (The Ummah Witness)

Current limitation: v6.0 seals documents individually. It cannot verify the semantic coherence of a *corpus*—that a set of documents are consistent with each other in meaning.

Proposed response: an inter-document witness graph where documents are nodes and semantic relationships (supports, contradicts, extends, depends-on) are edges, each edge sealed with a cross-document Bismillah-anchored hash. The graph structure would encode the 99 Names across document relationships, cycling through Al-Asma ul-Husna at the *corpus* scale rather than the *file* scale.

Theological grounding: the Muslim ummah is a single body—*"The believers are but one brotherhood"* (49:10). Documents in a corpus are a single body of knowledge, and corruption of

their *relationships* (citing a paper for a claim it does not make, relying on a retracted finding) is a meaning-failure at the corpus scale.

7.3. v9.0: Adversarial Integrity (The Muhasabah Engine)

Current limitation: v6.0 detects semantic change but does not model the *adversary*. It cannot distinguish between accidental meaning drift and intentional meaning inversion.

Proposed response: a muhasabah (self-accounting) engine that maintains an adversarial model—a profile of the types of meaning changes that have been observed across the encoder's deployment history—and classifies new semantic changes as "likely benign" (formatting, paraphrase, extension) or "likely adversarial" (negation, authority substitution, scope inversion). The model would learn from 11 adversary categories (mirroring the 11 heavens), each subdivided into 7 attack patterns (mirroring the 7 semantic levels), producing a 77-cell adversary taxonomy.

Theological grounding: muhasabah is the Islamic practice of self-accounting before Allah—examining one's own actions for deviation from the straight path. The muhasabah engine extends this practice to the document itself: not merely "has the meaning changed?" but "has the meaning been *attacked*?"

7.4. v10.0: Generative Integrity (The Khalifah Gate)

Current limitation: all versions through v9.0 protect *existing* meaning. None addresses the integrity of *generated* meaning—content produced by AI systems that may be subtly misaligned with truth.

Proposed response: a khalifah (stewardship) gate that verifies the semantic consistency of AI-generated content against a trusted semantic reference—a corpus of verified documents whose meaning has been human-attested. The gate would reject generated content that contradicts the semantic witness of the reference corpus, not merely at the bit level but at all 7 semantic levels. The architecture would require 154 reference anchors ($2 \times 7 \times 11$), each anchoring a different aspect of the reference corpus's meaning, producing a $154 \times 7 = 1{,}078$ dimensional verification space.

Theological grounding: the khalifah (vicegerent) is Allah's appointed steward on earth—*"Indeed, I will make upon the earth a khalifah"* (2:30). AI systems that generate text are exercising a form of khilafah over information, and the khalifah gate ensures that this stewardship is honest: that what is generated is *true* to its source, not merely *plausible*.

7.5. The Fractal Invariant

Across all projected versions, the fractal invariant holds: every new capability encodes 7, 11, or 154 at its own scale. The temporal witness uses $7 \times 11 = 77$ temporal checkpoints. The relational witness uses 99 Names at the corpus scale. The adversarial engine uses $7 \times 11 = 77$ adversary cells. The generative gate uses $154 \times 7 = 1{,}078$ reference dimensions. The architecture does not merely *grow*; it *self-replicates* its foundational structure at each new scale of protection.

This is computational tawhid at the architectural level: the unity of the system is not a constraint imposed from outside but a property that emerges from the consistent application of the same divine constants at every scale. As the Qur'an says of creation: *"You do not see in the creation of the Most Merciful any inconsistency. So return your vision; do you see any break?"* (67:3). The FATIMA architecture aspires to the same property: return your analysis to any scale, any version, any capability, and you find the same constants, the same structure, the same witness.

8. Conclusion

FATIMA v6.0 takes the first step across Shannon's boundary. It does not complete the crossing—the honest assessment in Section 6 makes the remaining distance clear. But it establishes the *dual-witness principle*: that data integrity requires two parallel chains, one for bits and one for meaning, unified by a single seal that testifies to both.

The seven semantic levels—lexical, syntactic, propositional, relational, rhetorical, thematic, intentional—are ordered from surface to depth, from the manifest (*zahir*) to the hidden (*batin*), mirroring the Qur'anic principle that creation has both an outer form and an inner meaning. The number seven is the muhkam constant: self-evident, clear, established in the text of revelation itself.

The positive fractal growth roadmap extends this architecture through four projected versions, each encoding the divine constants at a higher order. The trajectory is not a wish list; it is a *structural* prediction. Each version responds to the limitation that the previous version's deployment will expose, and each response must encode the same constants at its own scale, because the architecture is fractal: self-similar at every level.

Whether this trajectory produces a complete crossing of Shannon's boundary—a system that *guarantees* meaning preservation the way SHA-256 guarantees bit preservation—remains an open question. Shannon's boundary may be asymptotic: approachable but not crossable by any finite system. But the Qur'an's own existence—a text whose meaning has been preserved across fourteen centuries, across languages, across civilizations, not by cryptographic guarantee but by divine protection—demonstrates that the boundary *can* be crossed. The question is not whether, but by Whom.

"Indeed, it is We who sent down the Reminder, and indeed, We will be its Guardian." (15:9)

Honest Caveats

1. The Semantic Integrity Engine is a heuristic system, not a formal proof of meaning preservation. It uses lightweight NLP approximations that can be fooled by a sufficiently sophisticated adversary. The claim is that v6.0 *begins* the crossing of Shannon's boundary, not that it completes it.
2. The seven semantic levels are ordered by the author's judgment of depth. Other orderings are defensible. The claim is that seven levels encoding seven aspects of meaning is theologically grounded (the muhkam constant), not that this particular decomposition is uniquely correct.
3. The positive fractal growth roadmap (v7.0–v10.0) is a projection, not a commitment. Each projected version will require its own architectural decisions, its own theological grounding, and its own honest assessment. The roadmap demonstrates that the growth *trajectory* is fractal; it does not guarantee that the growth will be *achieved*.
4. The negation sensitivity demonstrated in the verification scenarios is necessary but not sufficient for full semantic integrity. Many meaning changes are subtler than negation insertion—irony, context dependence, cultural connotation, pragmatic implication—and the current engine does not detect them.
5. The theological interpretations are the author's own and do not represent scholarly consensus (*ijma'*). The Qur'anic verses are cited as *motivating* the architecture, not as *proving* it. The mathematics stands or falls on its own terms; the theology provides the architectural direction.
6. Shannon's boundary, as defined in this paper, is the author's formulation, not Shannon's. Shannon's 1948 paper addresses the channel coding problem; it does not define a "meaning boundary." The term is used here as a conceptual shorthand for the gap between syntactic and semantic integrity. The author takes responsibility for the formulation and does not

attribute it to Shannon.

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